from the Red Pitaya E1 header (FPGA)

E1 translators

Red Pitaya E1/E2 headers

4 x 74AVC4T774, 3.3 V <–> 1.8 V

33 R series, pull–ups/downs

File: dac_translators.kicad_sch

1.8 V quad SPI:
SCLK, CS_A/B_N, LDAC_N,
RESET_N, DQ_A0–3, DQ_B0–3,
ALERT_A/B_N

DAC digital

2 x AD3552R 16–bit current DAC

ADR4525 2.5 V reference

U1: ch 0, 1 U2: ch 2, 3

File: dac_digital.kicad_sch

per channel:
IOUTn to the TIA,
RFBn back from It,
VCMn

Analog channels

4 x ADA4807–1 TIA

feedback via the DAC RFB2

+/–5 V per channel

File: dac_analog.kicad_sch

OUTn_RFB
(TIA outputs)

Electrode outputs

4 x unity–gain inverter

49.9 R, BAV99 clamps, 1 M

8 x SMA (J4–J11), push–pull pairs

File: dac_output.kicad_sch

VLOGIC (+3V3 from E1)

AVDD, DVDD, VLOGIC (VREF from U7)

+/–5 V

+/–5 V

Power

+12 V in, polyfuse, TVS

AVDD LDO, +/–5 V pump, +2.5 V buck

+1.8 V LDO –> DVDD, VLOGIC

File: dac_power.kicad_sch

AD3552R 4–channel DAC mezzanine for the Red Pitaya STEMlab 125–14 4–Input

Four bipolar outputs, each a push–pull pair (OUTn+, OUTn–), +/–5 V per side, 16–bit, up to 9.7 V p–p differential.

Board: 95 x 60 mm, 6 layers (F.Cu / In1 GND / In2 / In3 / In4 power islands / B.Cu GND pour), mounts on E1 and E2.

Sockets J1/J12 must be elevated (>= 16.13 mm, e.g. Samtec ESW–113–33–T–D) to clear the Zynq heatsink.

Net names cross sheets through global labels; rails use power symbols.

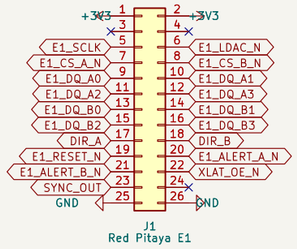
LQG Controller, CEL
Top level: signal flow and power distribution
CEL

Sheet: /
File: ad3552r_dac.kicad_sch

Title: AD3552R 4–channel DAC mezzanine (LQG Controller)

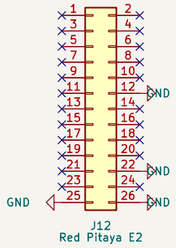
Size: A3	Date: 2026–11–02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/6

Red Pitaya E1 (3.3 V I/O, FPGA side)

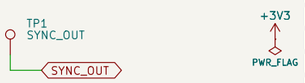


E1 pins 1–2 supply +3V3 (translator VCCA)

Red Pitaya E2 (ground and support only)

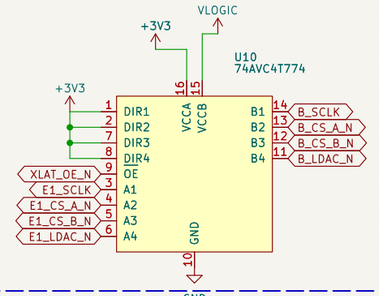


Test point

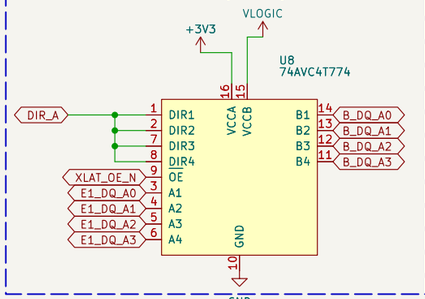


SYNC_OUT (E1 pin 23): FPGA marker for scope triggering

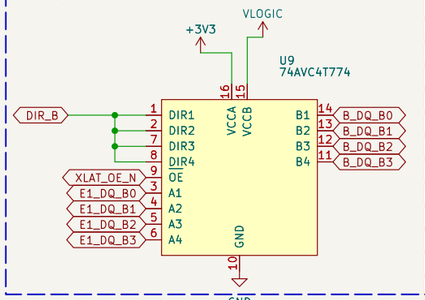
U10: SCLK, CS_A/B_N, LDAC_N (A->B)



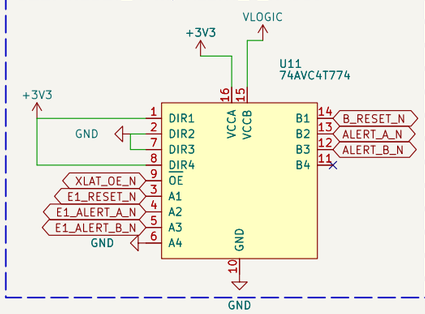
U8: quad-SPI lane A (DIR_A)



U9: quad-SPI lane B (DIR_B)

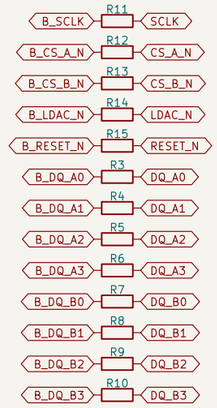


U11: RESET_N in, ALERT_x_N out



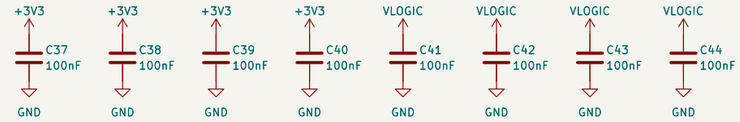
-OE (XLAT_OE_N) is driven by the FPGA and pulled up by R16: outputs stay off until the bitstream runs

33 R series (DAC side)



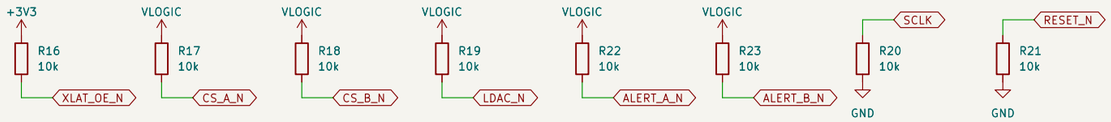
all 33 R, 0402

Translator decoupling (100 nF at each VCCA and VCCB pin)



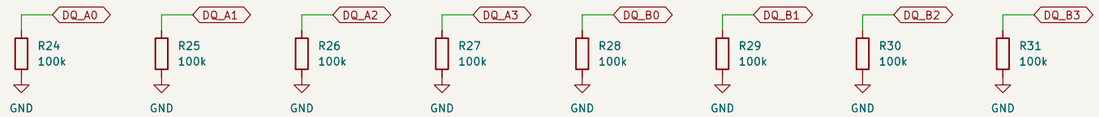
C37–C40 VCCA (+3V3), C41–C44 VCCB (VLOGIC)

Pull-ups and pull-downs: defined levels while the FPGA is not configured



10 k: OE high (translators off), CS/LDAC high, ALERT pulled up, SCLK low, RESET_N low (DACs held in reset)

100 k pull-downs on the DAC data lines



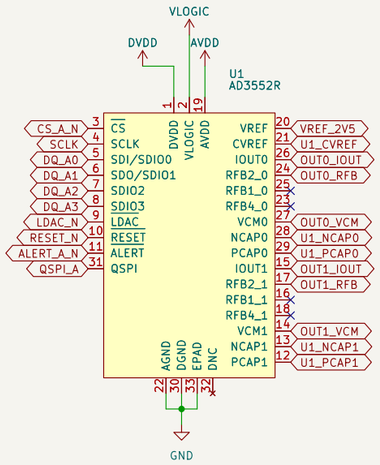
LQG Controller, CEL
E1 3.3 V (VCCA) <-> DAC 1.8 V (VCCB = VLOGIC); 33 R series at the DAC side
CEL

Sheet: /E1 translators/
File: dac_translators.kicad_sch

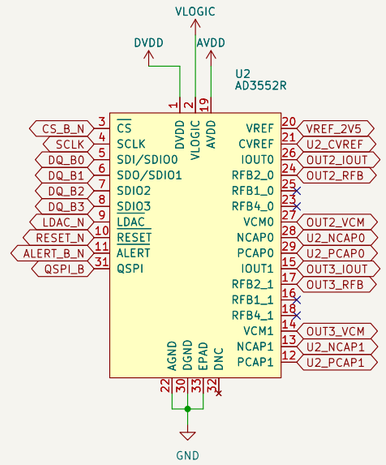
Title: AD3552R DAC mezzanine: Red Pitaya interface and level translators

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/6

U1: AD3552R, channels 0 and 1

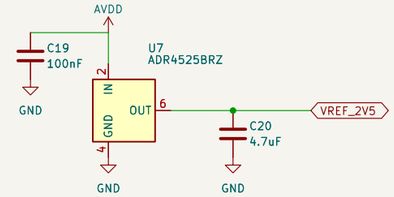


U2: AD3552R, channels 2 and 3



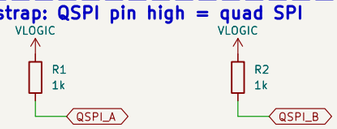
Range: RFB2 path, output span +/−5 V (register 0x19 = 0x33). Unused RFB1/RFB4 pins left open.

Voltage reference: ADR4525, 2.500 V

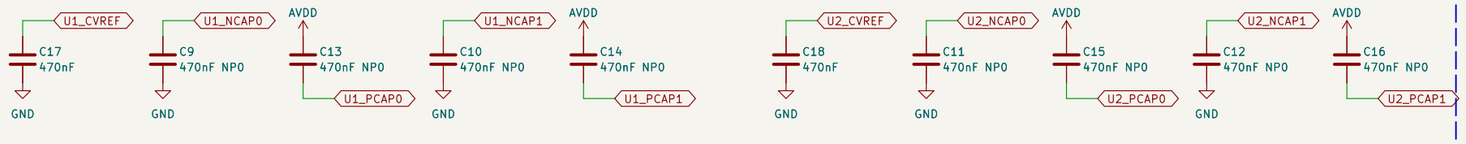


External−reference mode: VREF_2V5 feeds both DACs; C20 4.7 uF at the output

Interface strap: QSPI pin high = quad SPI



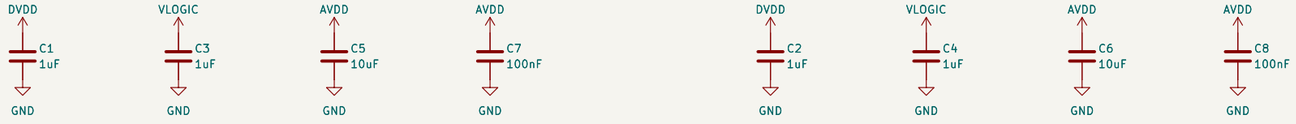
DAC internal−reference and output−stage capacitors (placed at the DAC pins)



U1 **U2**

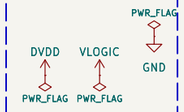
470 nF NP0 on NCAP/PCAP (output−stage compensation), 470 nF on CVREF (internal reference)

DAC supply decoupling (one set per DAC at its pins)



DVDD 1 uF, VLOGIC 1 uF, AVDD 10 uF + 100 nF

Flags



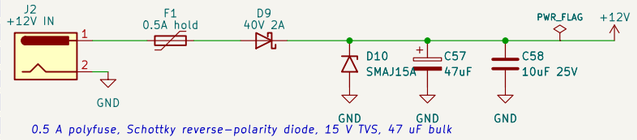
LQG Controller, CEL
U1 −> OUT0/OUT1, U2 −> OUT2/OUT3; shared SCLK, LDAC_N, RESET_N
CEL

Sheet: /DAC digital/
File: dac_digital.kicad_sch

Title: AD3552R DAC mezzanine: DACs and voltage reference

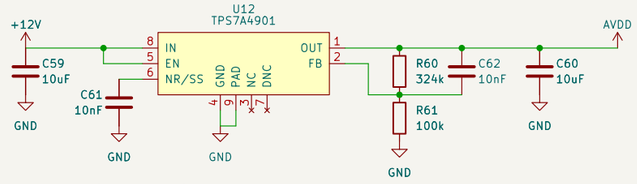
Size: A3	Date: 2026−11−02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/6

+12 V input and protection



0.5 A polyfuse, Schottky reverse-polarity diode, 15 V TVS, 47 uF bulk

AVDD 5.02 V: low-noise LDO (TPS7A4901)



AVDD = V_FB (1 + R60/R61); C62 feed-forward; C61 soft start / noise filter

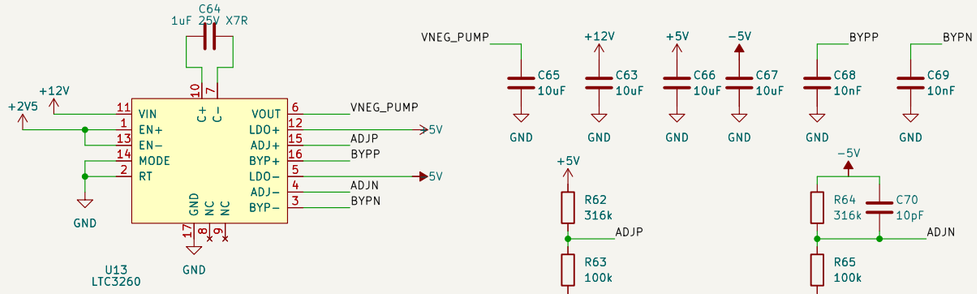
Rails and start-up

+12V	external, about 145 mA total (1.7 W)
AVDD	5.02 V, DAC analog supply and ADR4525 input
+5V	LTC3260 LDO+, amplifier positive rail
−5V	LTC3260 LDO−, amplifier negative rail
+2V5	AP63201 buck: feeds the 1.8 V LDO, enables U13
+1V8	TLV75518P -> DVDD (FB1) and VLOGIC (FB2)
+3V3	from the Red Pitaya E1 header (VCCA)

Start-up order (the AD3552R wants AVDD first):

1. +12 V on: U12 brings up AVDD
2. AVDD above about 1.4 V enables U14: +2V5
3. +2V5 enables U15 (+1V8) and U13 (+/−5 V)
4. RESET_N held low by R21 until the FPGA releases it

+/−5 V: inverting charge pump with positive and negative LDOs (LTC3260)

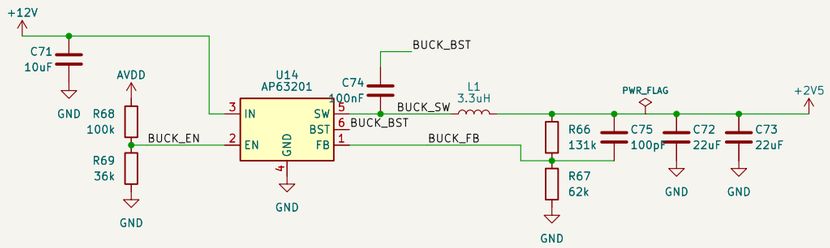


EN+/EN− from +2V5: the pump starts after the buck

+5 V = 1.2 V (1 + R62/R63)

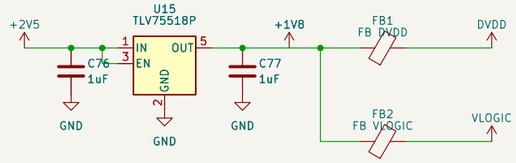
−5 V = −1.2 V (1 + R64/R65); C70 feed-forward

+2.5 V: buck converter (AP63201), enabled once AVDD is up



EN = AVDD x R69/(R68+R69): starts after AVDD; +2.5 V = 0.8 V (1 + R66/R67)

+1.8 V LDO (TLV75518P) -> DVDD and VLOGIC



Separate beads keep the DAC digital current off the logic supply

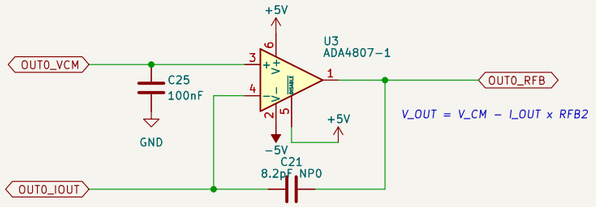
LQG Controller, CEL
+12 V in -> AVDD 5.02 V, +/−5 V, +2.5 V, +1.8 V (DVDD, VLOGIC)
CEL

Sheet: /Power/
File: dac_power.kicad_sch

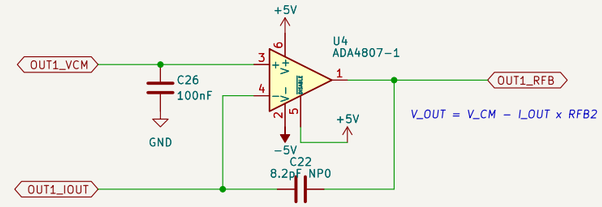
Title: AD3552R DAC mezzanine: power supplies

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/6

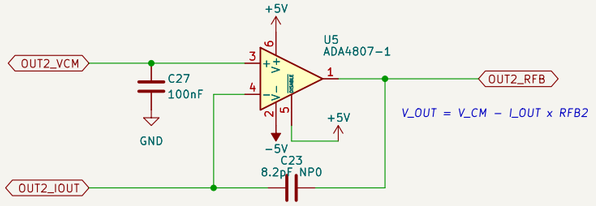
Channel 0 transimpedance amplifier (U1 IOUT0 / RFB2_0)



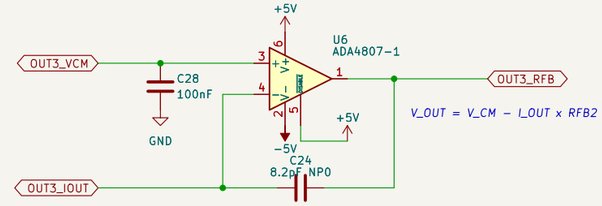
Channel 1 transimpedance amplifier (U1 IOUT1 / RFB2_1)



Channel 2 transimpedance amplifier (U2 IOUT0 / RFB2_0)



Channel 3 transimpedance amplifier (U2 IOUT1 / RFB2_1)



Supply decoupling: one 100 nF pair at each amplifier



The feedback resistor RFB2 (3.228 k) sits inside the AD3552R between IOUTx and RFBx: each amplifier output goes back to the DAC RFB pin. C_F (8.2 pF NP0) across it sets the bandwidth; VCM from the DAC biases the + input.

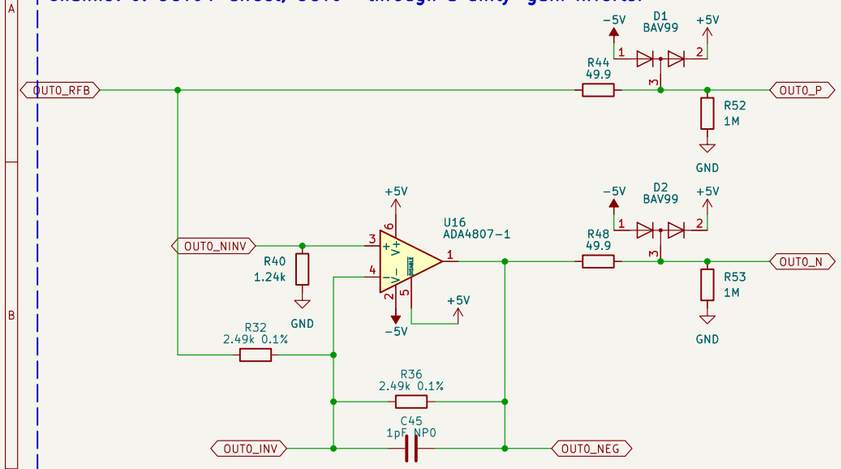
LQG Controller, CEL
IOUT -> voltage: RFB2 (3.228 k) is inside the AD3552R; C_F 8.2 pF NP0
CEL

Sheet: /Analog channels/
File: dac_analog.kicad_sch

Title: AD3552R DAC mezzanine: transimpedance amplifiers

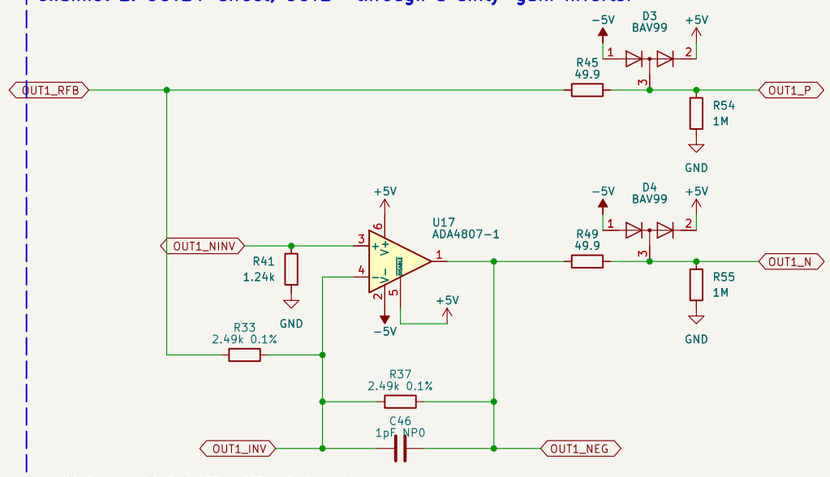
Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/6

Channel 0: OUT0+ direct, OUT0− through a unity-gain inverter



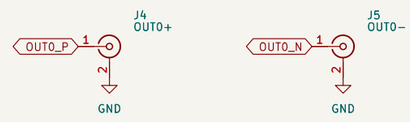
V(OUT−) = −V(OUT+): R32 = R36 = 2.49 k 0.1%

Channel 1: OUT1+ direct, OUT1− through a unity-gain inverter

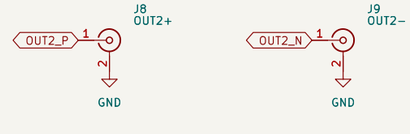


V(OUT−) = −V(OUT+): R33 = R37 = 2.49 k 0.1%

SMA outputs (Amphenol 132134)



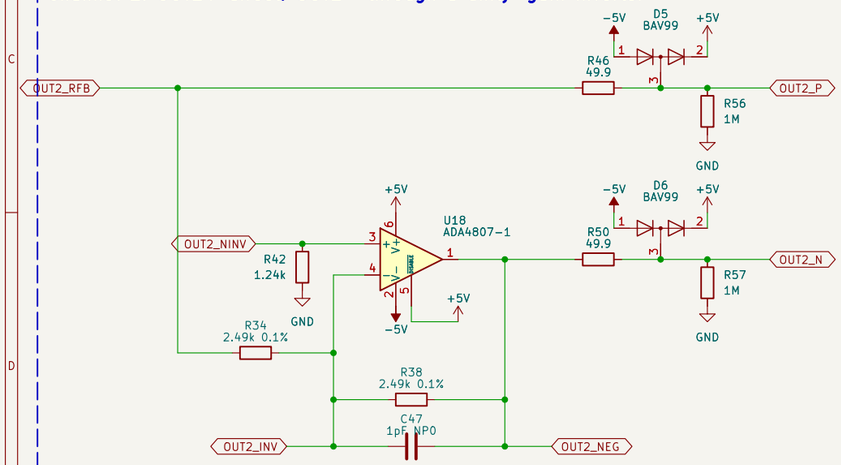






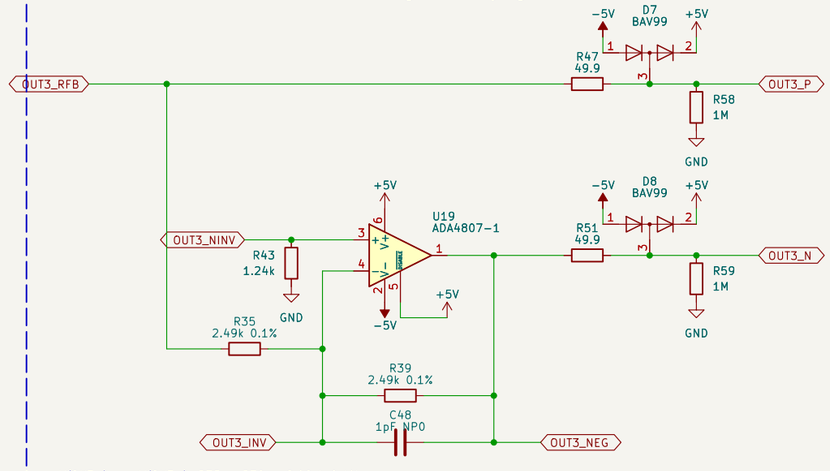
Top edge: OUT0, OUT1 Bottom edge: OUT2, OUT3

Channel 2: OUT2+ direct, OUT2− through a unity-gain inverter



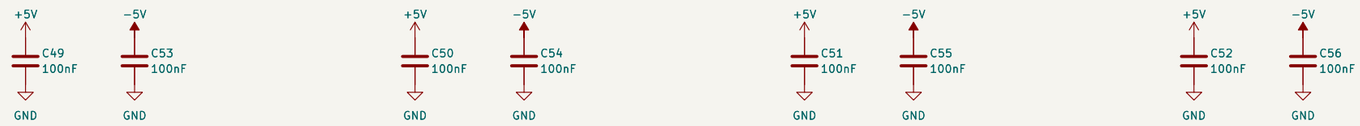
V(OUT−) = −V(OUT+): R34 = R38 = 2.49 k 0.1%

Channel 3: OUT3+ direct, OUT3− through a unity-gain inverter



V(OUT−) = −V(OUT+): R35 = R39 = 2.49 k 0.1%

Inverter supply decoupling: one 100 nF pair at each amplifier

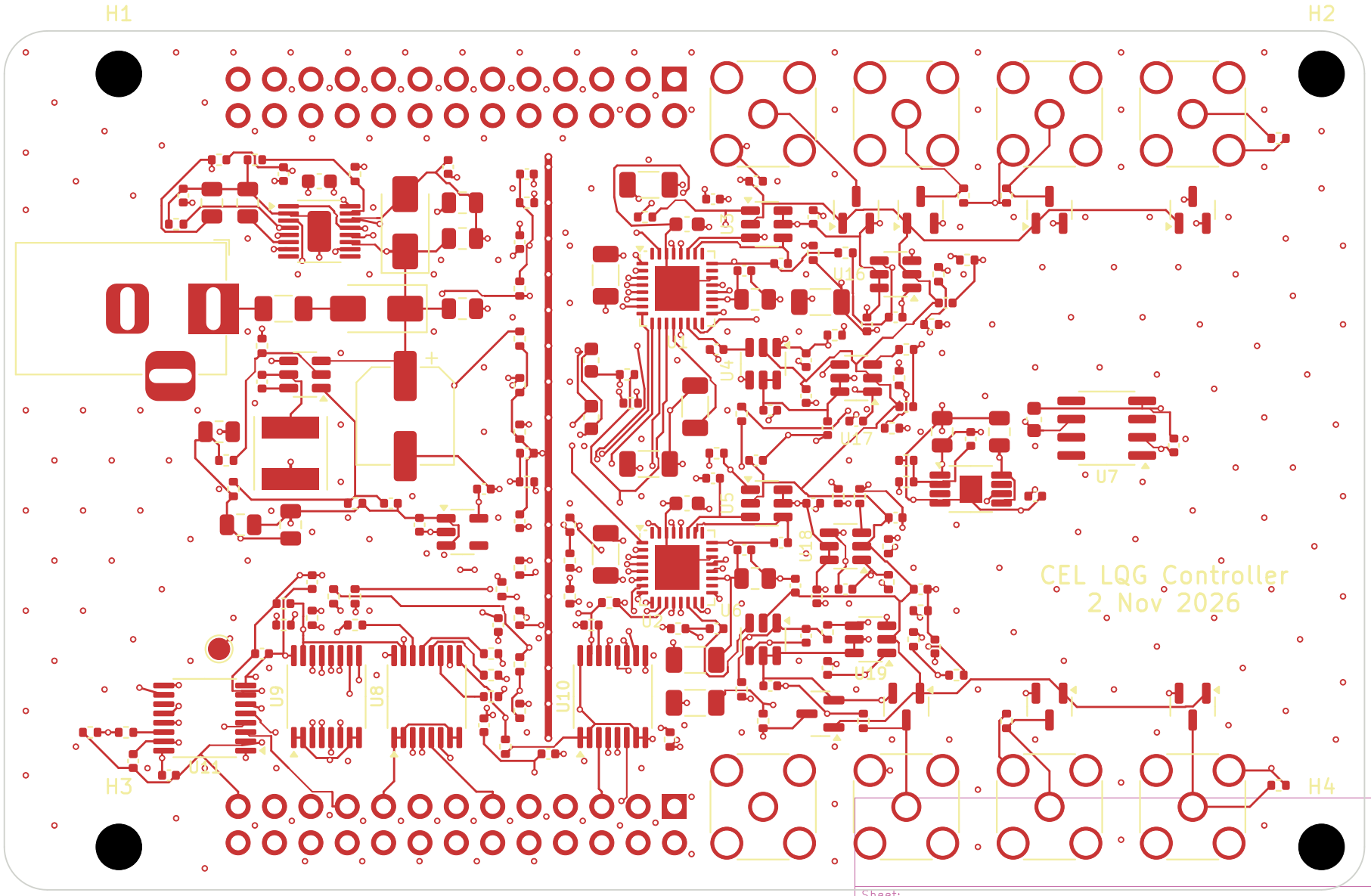


LQG Controller, CEL
OUTn+ = TIA output, OUTn− = inverted copy: up to 9.7 V p−p differential per channel
CEL

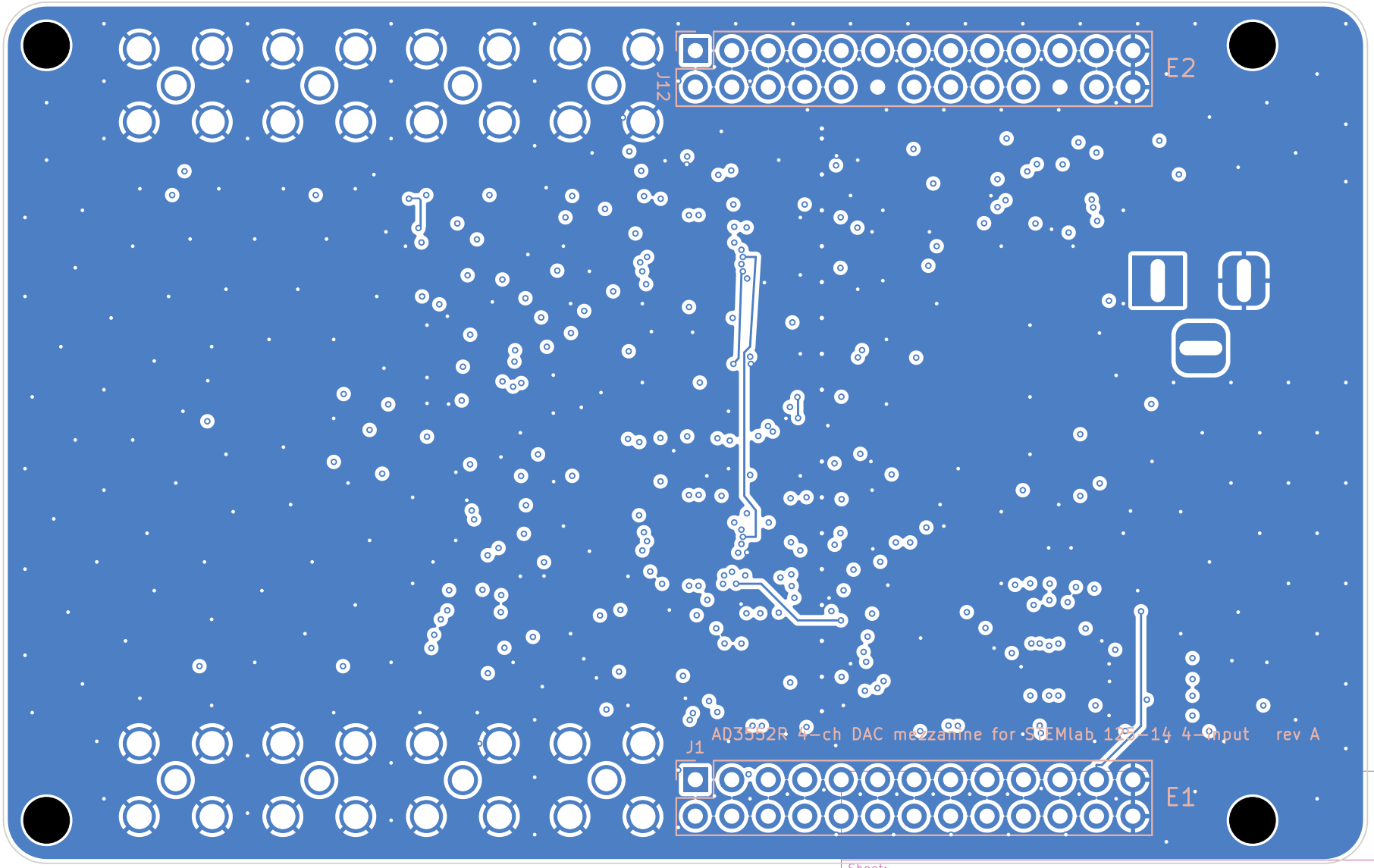
Sheet: /Electrode outputs/
File: dac_output.kicad_sch

Title: AD3552R DAC mezzanine: push−pull electrode outputs

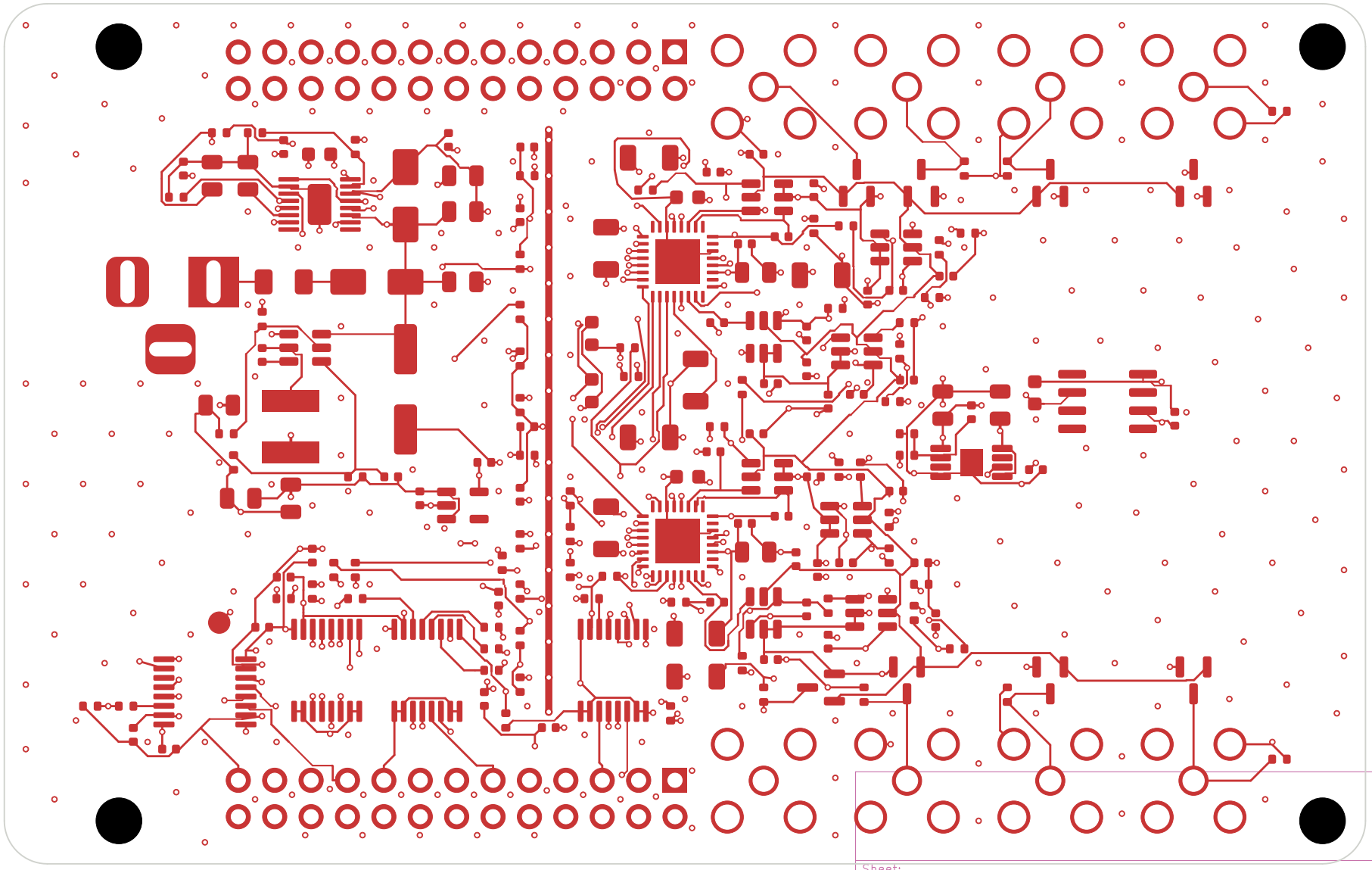
Size: A3	Date: 2026−11−02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/6



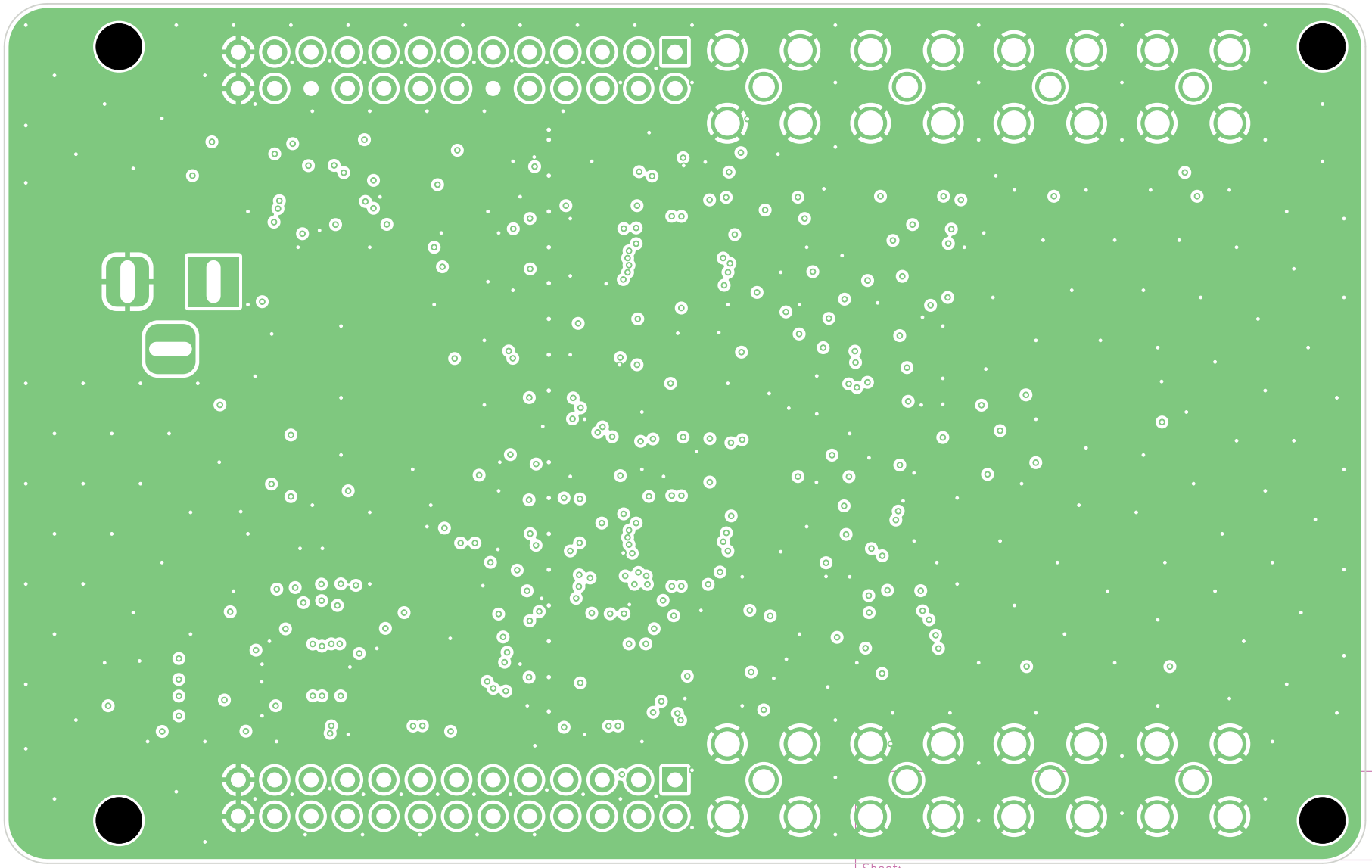
Sheet:		
File: ad3552r_dac.kicad_pcb		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/1



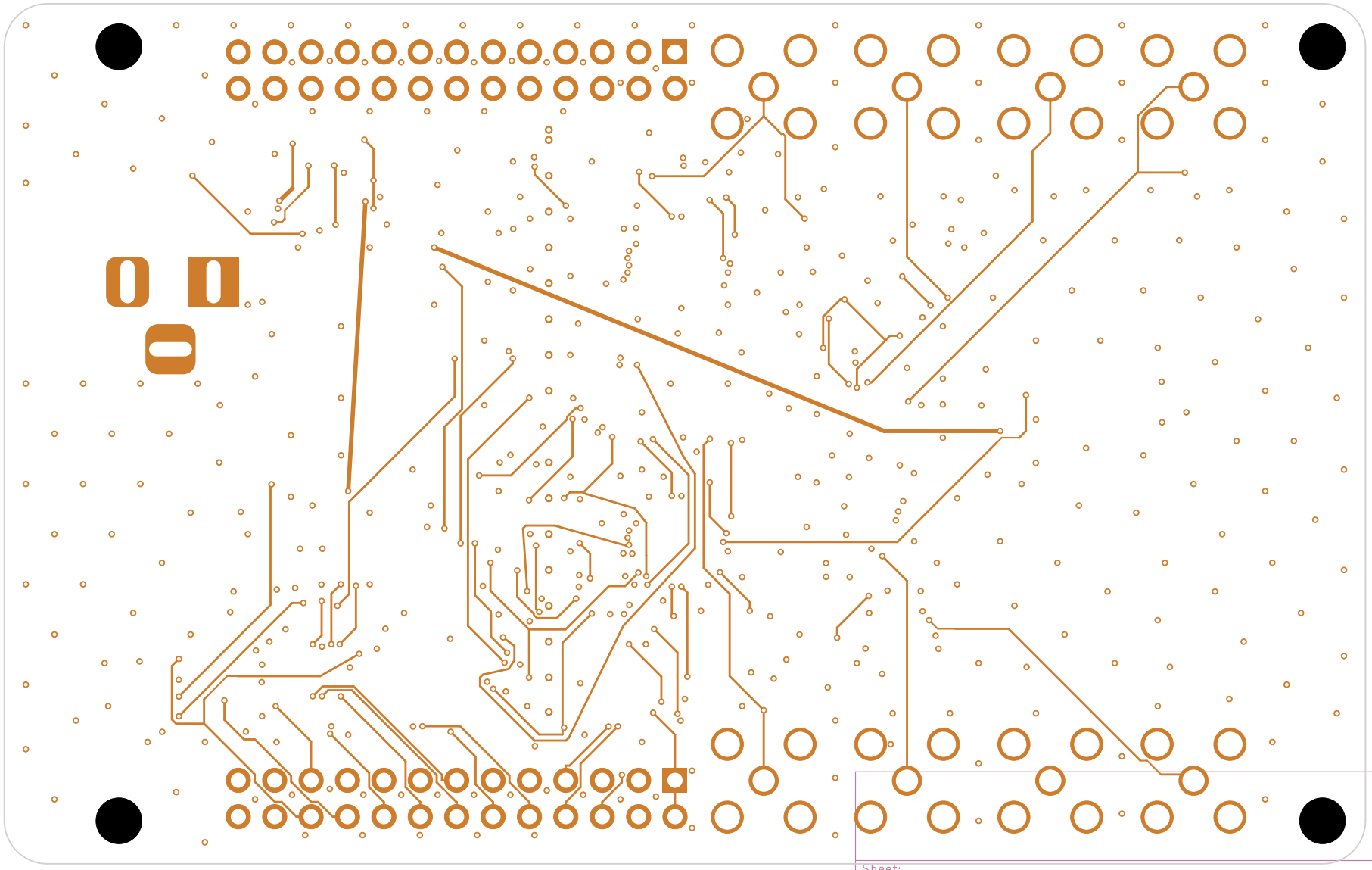
Sheet:		
File: ad3552r_dac.kicad_pcb		
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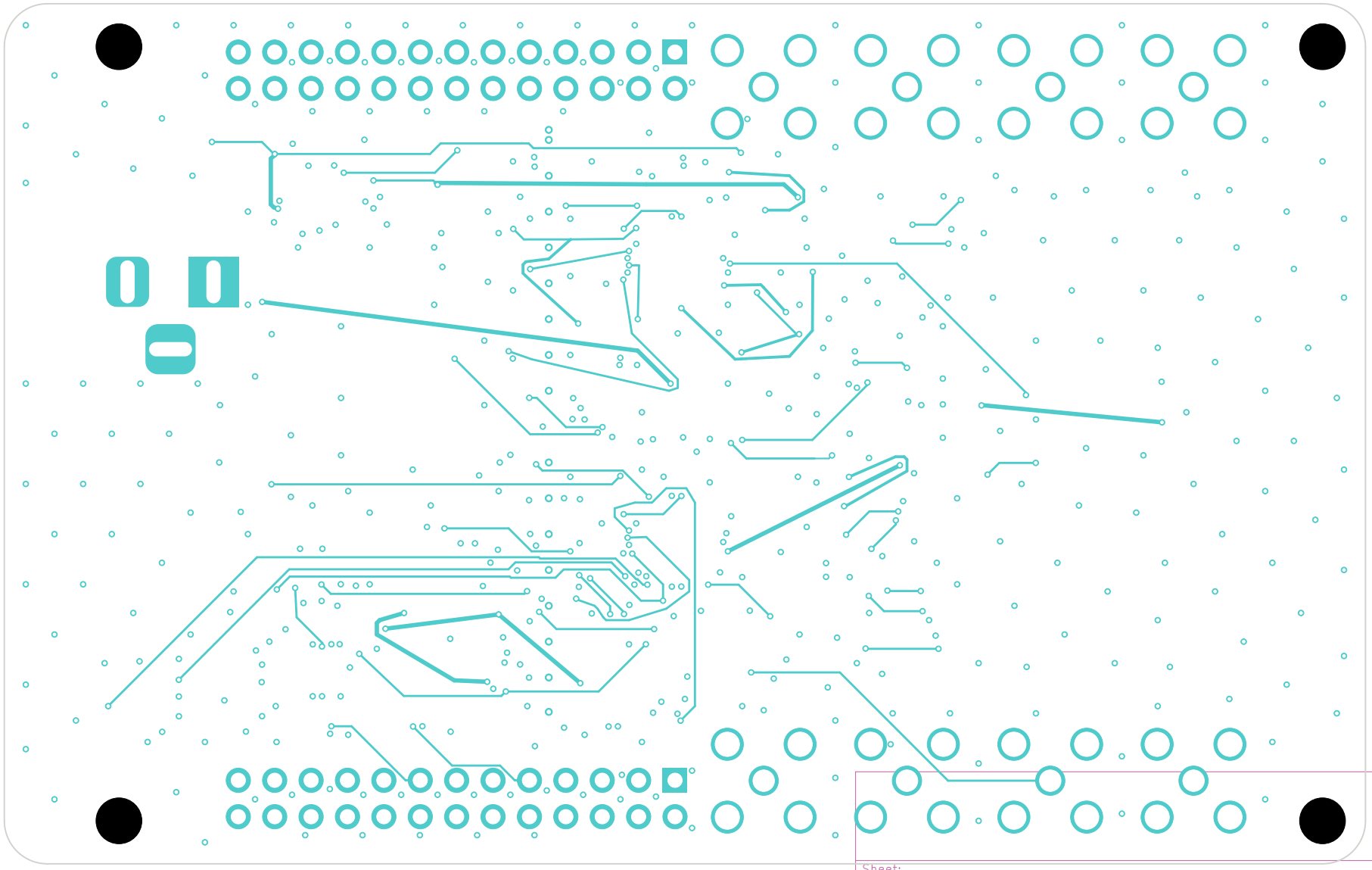
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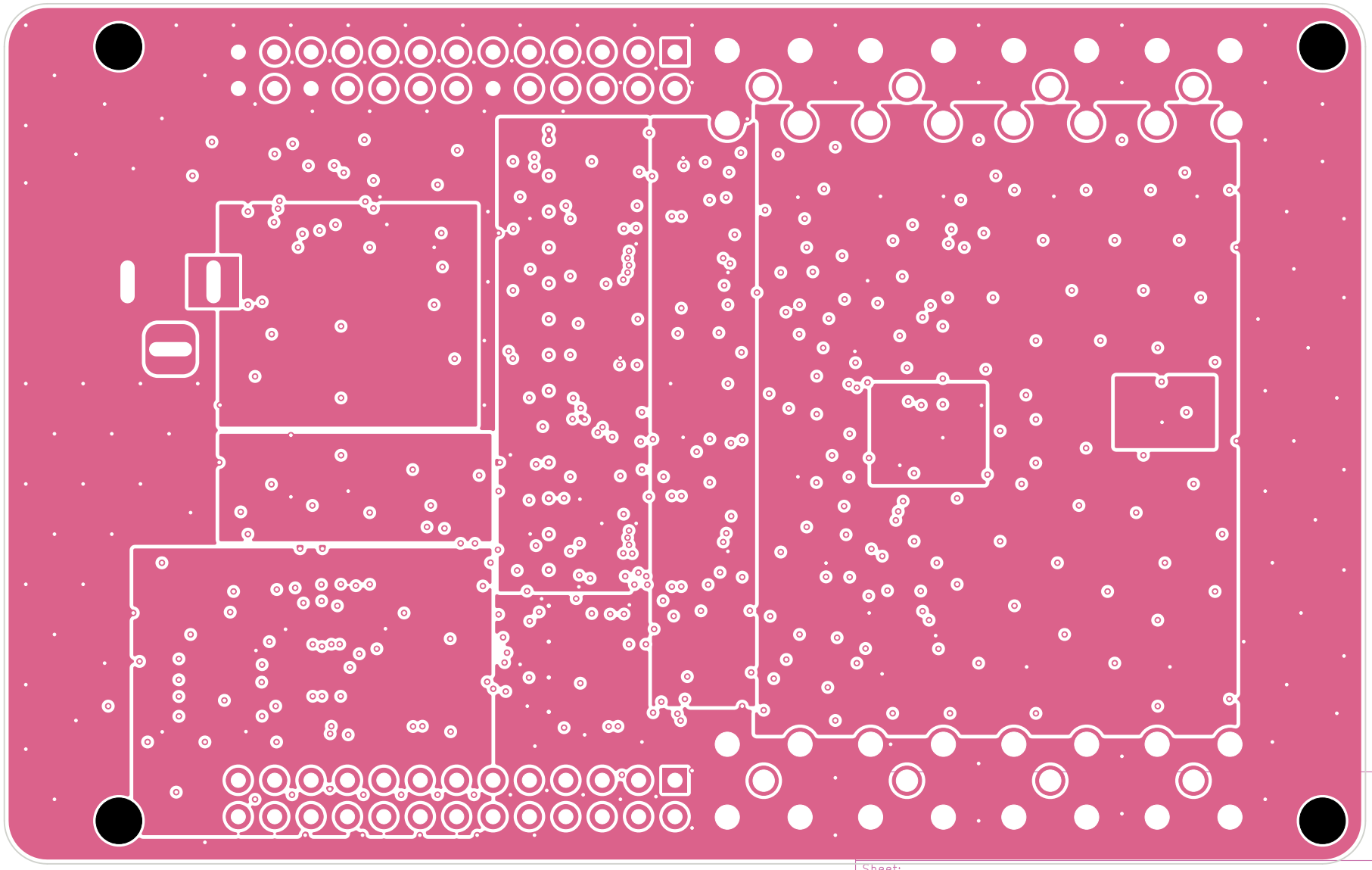
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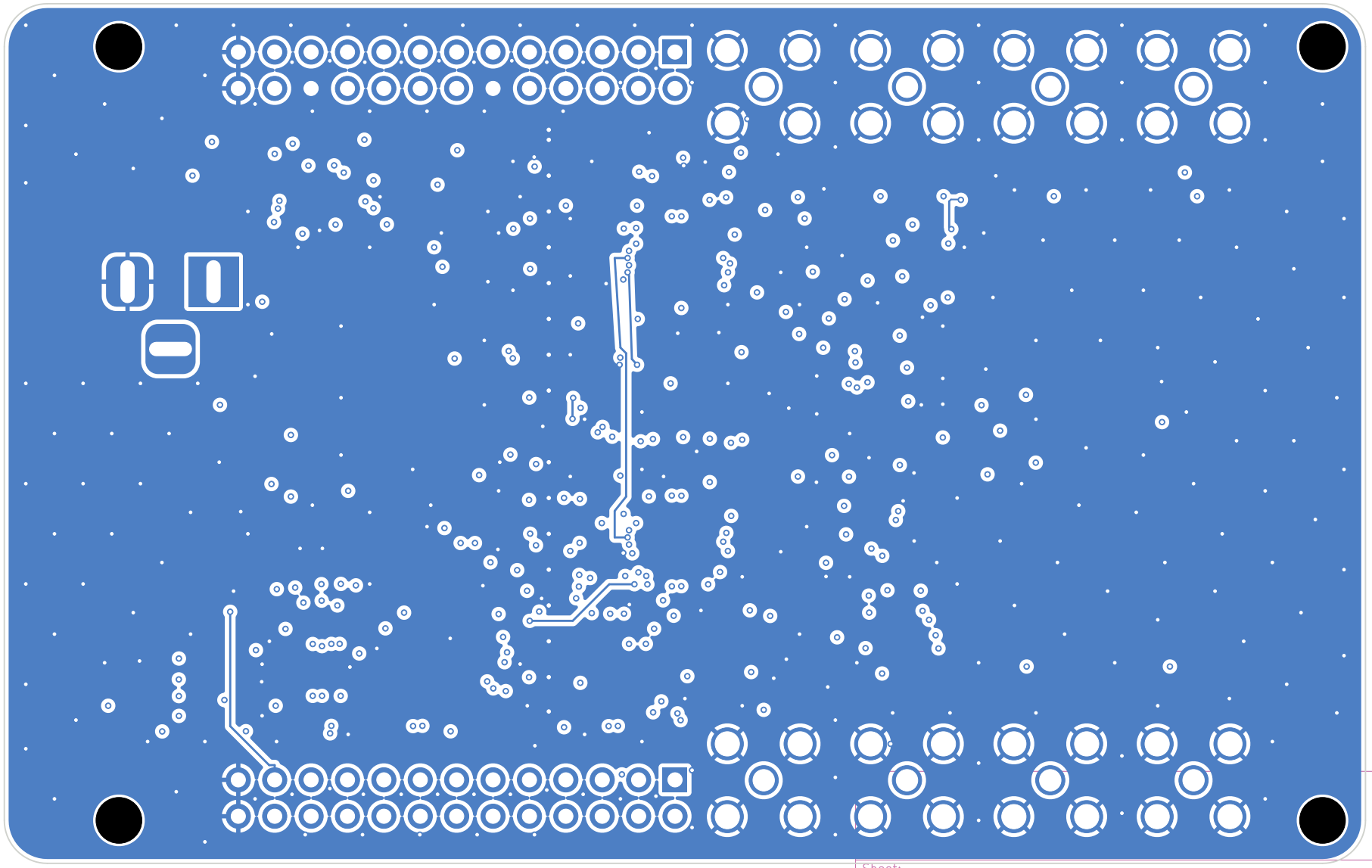
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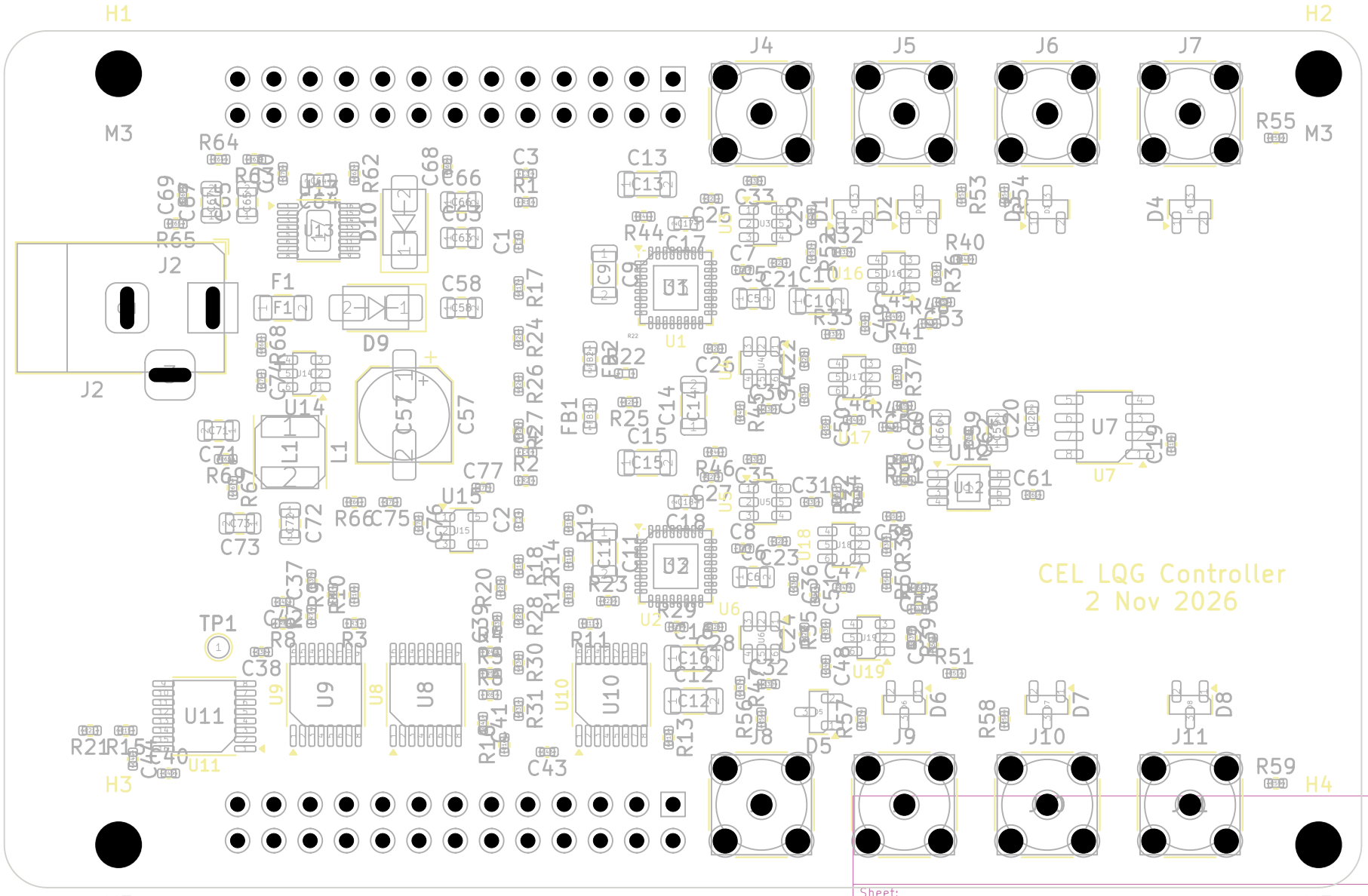
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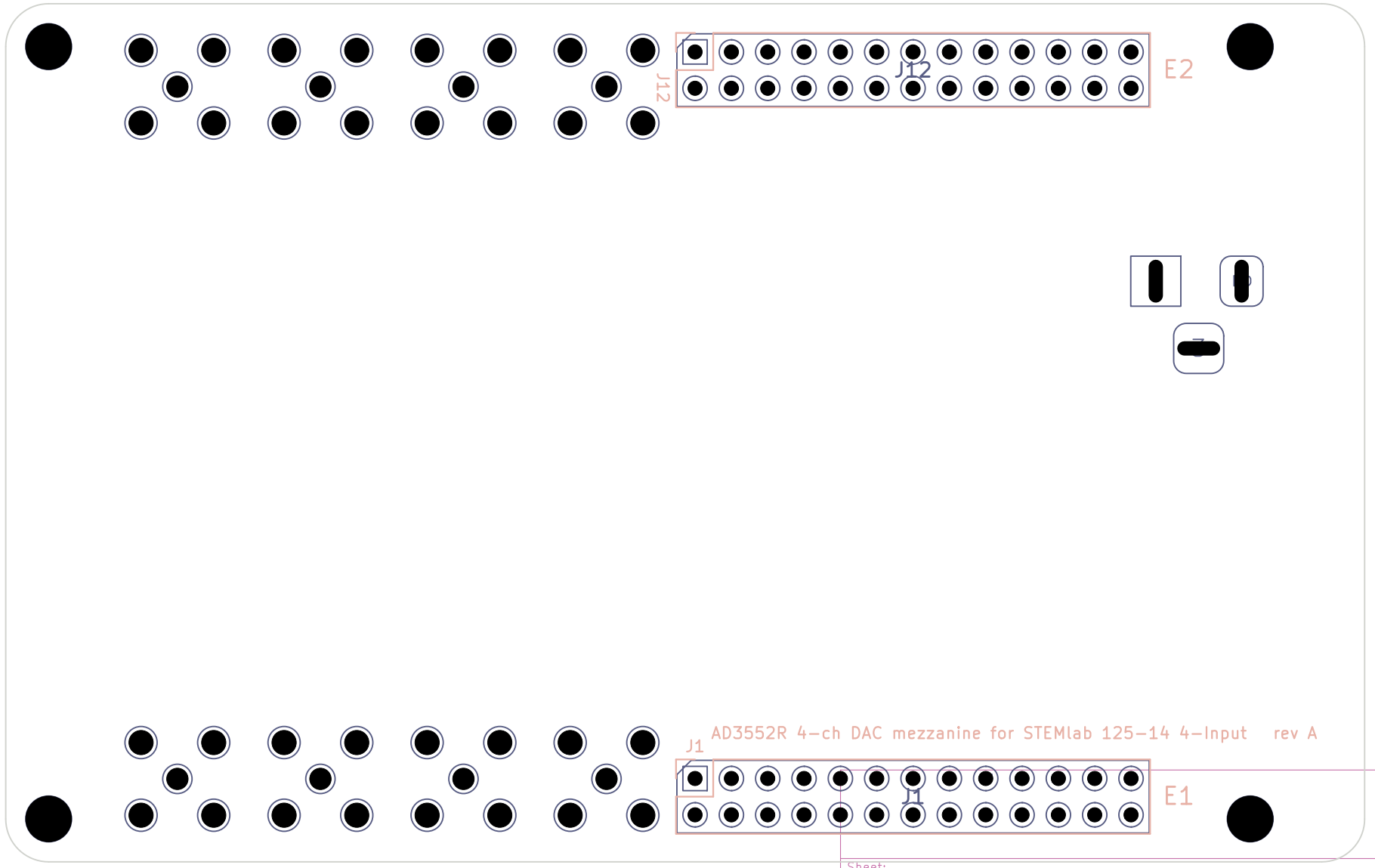
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